

Ethical issues around data & information

Data isn't just numbers in a spreadsheet — it's people. This dotpoint is about the line between what you can do with data and what you should do.

What this key knowledge covers

This summary distils everything you need for **KK06 — Ethical issues around data & information** in Data Analysis. Work through the explanations, then use the worked good-vs-weak answers to see exactly how marks are won and lost. Tick off the revision checklist at the end before a SAC or exam.

Study summary

Data vs information — quick reset

Data is the raw stuff you collect (a student ticks "3 hours sleep"). **Information** is what you get after analysis gives it meaning ("Year 11 students average 5.8 hours sleep on school nights — well below the recommended 8–10"). Ethical issues live at *both* ends: how you **gathered** the raw data, and what you **do with** the information you produce.

Legal is the floor. Ethical is the ceiling.

This is the single idea examiners reward most — and the one weak answers miss. **Legal** means "the law allows it." **Ethical** means "it's the right thing to do." They overlap, but they are *not* the same question.

In Unit 1 you study the **Australian Privacy Principles** (KK05) — that's the *law*. KK06 sits one level up: even when the Privacy Act permits something, you still have to ask "*should we?*" Use this 2×2 to keep them apart:

You ask for consent, explain the purpose, anonymise results, store data securely and delete it when done.

The fine print technically lets you reuse the wellbeing data to rank students — but it's not what anyone expected or agreed to.

Where most exam questions live.

Leaking that the data was being secretly sold, to protect students — breaks a rule, but defends people.

Selling identifiable student mental-health data to an advertiser without consent.

If a question uses the word **ethical**, do *not* answer "because it breaks the Privacy Act." That's a **legal** answer. Talk about fairness, honesty, harm to people, trust and consent — the *should*, not



the *allowed*.

Five ways to interrogate any dataset

Think of each ethical issue as a **lens** you hold up to a dataset. The same PulseCheck spreadsheet looks different through each one. Tap a lens below (or click a ring in the scene) to focus it.

Can people see what you collect, why, and how you reached your conclusions? Hidden methods and misleading charts fail this lens — even if every number is true.

Transparency

Transparency means people can **see and understand** what's happening to their data: what you're collecting, the purpose, where it's stored, who sees it, and *how* your analysis turned their answers into a conclusion. The opposite is a "black box" — data goes in, claims come out, and no one can check the working.

For a data analyst, transparency also covers your **visualisations**. A chart with a chopped-off axis or a cherry-picked date range can be technically "true" yet deeply misleading. Hiding your method is a transparency failure even if every number is real.

Privacy is about *keeping data from the wrong eyes*. Transparency is about *being honest with the right people* — the ones whose data it is. You can be private and still sneaky.

Inaccurate or incomplete data

There's a quality side to this (that's KK03) and an **ethical** side (that's KK06). The ethical problem is what happens when you *present shaky data as if it were solid*, and real people get judged or affected by a conclusion that the data can't actually support.

Inaccurate = the values are wrong (typos, jokes, lies, broken sensors). **Incomplete** = there are holes (people skipped questions, only some groups replied, the sample is too small to represent everyone). Either way, the *information* you build on top is untrustworthy — but a slick chart can hide that completely.

Ownership and control of data & information

Ownership asks: **whose data is this?** Control asks: **who gets to decide what happens to it?** These come apart all the time. The student *feels* their survey answer is theirs — but the organisation that collected and stored it may be the one holding the keys. That tension is the ethical issue.

Good practice gives people meaningful control: the right to see their own data, correct it, withdraw it, and have it deleted. The questions to ask: *Can a participant pull their data back out? Who keeps the dataset when the project ends? Can it be handed to someone else?*

Misuse of personal data

Personal data is anything that can identify a person or reveal something private about them. **Misuse** is using it beyond the purpose it was given for — sometimes called *function creep*, where data quietly slides from its original job into a new one no one signed up for.

Misuse includes: profiling and labelling people, surveillance, singling individuals out, combining datasets to re-identify "anonymous" people, and disclosing or selling data. The harm is the point

— misuse erodes trust and can hurt the very people who shared their data in good faith.

Stripping names off data feels safe, but combining it with other details (timestamps, class, age, suburb) can **re-identify** people. "Anonymous" data is only anonymous until someone joins it to something else.

Repurposing & sharing of data by AI

When you paste data into an AI chatbot or generator, you may be **handing it to a third party**. That data can be stored on their servers, used to **train** future models, and effectively shared beyond your control — repurposed for something completely different from your project.

AI also *repurposes* data in a second sense: it can combine sources, infer new facts about people, and re-identify individuals — generating conclusions people never consented to. The convenience of "just ask the AI to summarise it" hides a one-way door: once personal data leaves your hands, you can't pull it back.

Transparency · **A**ccuracy & completeness · **O**wnership & control · **M**isuse · **A**I repurposing & sharing. A memory hook: *"TAOMA — Tell the truth, About the data, Own it together, Misuse never, AI with care."*

Ethics is about who gets hurt

The strongest exam answers don't just *name* an issue — they trace the **harm to real people**. In the centre is a PulseCheck decision. Around it sit the people affected. Trigger an ethical lapse and watch the ripple reach them.

"It's a misuse of data" = 1 mark, maybe. "It's a misuse of data **because** students shared mental-health answers to get help, and instead they're individually flagged to staff — eroding trust and discouraging honest responses in future" = full marks. Always follow the ripple to a person.

Which lens is this?

Read each PulseCheck moment and pick the ethical issue it best illustrates. Some are sneaky — they could touch more than one, but choose the *primary* one.

Flip cards

Define it in your head first, then tap to check. These are the exact phrasings to have ready in the exam.

Where marks quietly vanish

Every one of these comes straight from how this dotpoint is actually marked. Read them like a checklist before you write.

The P·E·L method for ethics answers

Ethics questions reward structure. Name it, explain the harm, tie it back to the case. Hit all three and you cover what the marking guide looks for.

Command words decoder

The verb tells you how much to write. Match your effort to the marks.

VCE-style questions

Six questions, 26 marks. Try each before revealing. Use ☐ to see a common weak answer, ☐ for a strong one, and ☐ for the marking guide — then self-mark to feed the score dock.

A local gym, **StepUp**, gives members a free fitness-tracker app. Members log workouts, weight, heart rate and sleep. StepUp's analyst studies the data to design class timetables. Recently, management asked the analyst to (a) email each member a "fitness score" ranking them against others, and (b) paste the member data into an AI tool to auto-generate personalised marketing. Just like the exam, the issues are the same — only the scenario is new.

- 1 mark: correctly names **misuse of personal data** (function creep / using data beyond its agreed purpose).
- 1 mark: identifies the affected group — the **members** whose data is repurposed.
- No marks for citing a law — the question asks for the ethical issue, not legality.
- 1 mark: names the issue — **AI repurposing and sharing** of data.
- 1 mark: explains the mechanism — data may be stored, train models, or be shared with a third party beyond the analyst's control.
- 1 mark: links to harm/consent — members' sensitive health data is exposed without their agreement.
- Transparency (2): clear statement of what/why/where/who BEFORE collection so members can give informed consent.
- Ownership & control (2): members can access, correct, withdraw and delete their data; it isn't handed on without consent.
- Answers must be specific to StepUp's fitness data, not generic "be honest".
- Identifies at least two relevant issues — typically inaccurate/incomplete data and transparency (misuse also creditable).
- For each: explains WHY it's an ethical issue (not just a quality label) and the impact on people (members, public).
- Shows the tension/both sides expected by "discuss" — e.g. business's right to promote vs duty to be honest.
- Reaches a reasoned position. 5–6 marks: two issues fully developed with stakeholders and a justified stance.
- Addresses all five lenses with a concrete action for each (transparency, accuracy, ownership, misuse, AI).
- Makes a clear recommendation rather than a list of generic verbs.
- Justifies WHY — ties back to member trust, consent and avoiding harm to people/business.
- 4–5 marks: coherent policy across all five with genuine justification, not just restating the lenses.

Common traps & misconceptions

Exam trap

TRAP 01 Answering "ethical" with a law Writing "it breaches the Privacy Act" when the question asks about an ethical issue. That's a legal answer to an ethics question. **Fix:** Talk about fairness, honesty, harm, trust and consent — the should, not the allowed.

Exam trap

TRAP 02 Naming an issue with no harm "This is a transparency issue." Full stop. A bare label with no explanation or stakeholder barely scores. **Fix:** Always follow with WHY it matters and WHO is affected — the P·E·L spine.

Exam trap

TRAP 03 Forgetting the AI issue Listing only four issues. The repurposing and sharing of data by AI is examinable and the one most often left out. **Fix:** Memorise all five: Transparency, Accuracy, Ownership, Misuse, AI.

Exam trap

TRAP 04 Confusing transparency with privacy Treating them as the same. Privacy is keeping data from the wrong eyes; transparency is being honest with the right people. **Fix:** You can be private and still sneaky. Different lens.

Exam trap

TRAP 05 Mixing up KK03 quality with KK06 ethics Listing data-quality factors (accuracy, bias, integrity) instead of the ethical impact of acting on bad data. **Fix:** KK06 = the harm caused when shaky data drives a real decision.

Exam trap

TRAP 06 Saying "anonymous = safe" Assuming stripping names makes data harmless. Combining it with other details can re-identify people. **Fix:** Note re-identification risk — anonymisation is not a free pass.

Exam trap

TRAP 07 Vague "it's wrong / bad" Emotional verdicts with no reasoning. Markers can't award "it's just bad." **Fix:** State the specific issue and trace a concrete consequence to a named stakeholder.

How to answer exam questions

Read the **command word** first — it sets how much to write. *State/Identify* wants a word or phrase; *Describe* wants features; *Explain* wants reasons and links; *Justify/Evaluate* wants a judgement backed by evidence. Always pull in the **scenario detail** rather than answering in the abstract. The examples below contrast a weak response with a strong one for the same question.

Q1. Identify · 2 marks

StepUp asks its analyst to email every member a "fitness score" that ranks them against all other members. **Identify** the ethical issue this raises and the group of people most affected. ()

✗ A weak answer

It's bad to rank people and it's probably against privacy law.

✓ A strong answer

This is a **misuse of personal data**: fitness data members gave to plan classes is repurposed to rank and compare them. The people most affected are the **StepUp members**, who never consented to being publicly scored against others.

How the marks are awarded

- 1 mark: correctly names **misuse of personal data** (function creep / using data beyond its agreed purpose).
- 1 mark: identifies the affected group — the **members** whose data is repurposed.
- No marks for citing a law — the question asks for the ethical issue, not legality.

Q2. Explain · 3 marks

Management tells the analyst to paste members' workout, weight, heart-rate and sleep data into a free public AI tool to auto-generate marketing. **Explain** one ethical issue with doing this. ()

✗ A weak answer

The AI might get it wrong so the marketing could be inaccurate. That would look unprofessional.

✓ A strong answer

This raises the **repurposing and sharing of data by AI**. Pasting members' personal health data into a public AI tool may store it on a third party's servers and use it to train future models — sharing it well beyond StepUp's control. Members shared this data to improve their gym experience, not to have sensitive health information leave the business and be reused without their consent.

How the marks are awarded

- 1 mark: names the issue — **AI repurposing and sharing** of data.
- 1 mark: explains the mechanism — data may be stored, train models, or be shared with a third party beyond the analyst's control.
- 1 mark: links to harm/consent — members' sensitive health data is exposed without their agreement.

Q3. Describe · 4 marks

Describe how **transparency** and **ownership/control** could each be respected when StepUp collects and uses member fitness data. ()

✗ A weak answer

Be honest with members and let them own their data. Don't do anything sneaky.

✓ A strong answer

Transparency: StepUp tells members up front exactly what is collected (workouts, weight, heart rate, sleep), why (to design class timetables), where it's stored and who sees it — so members can make an informed choice to join. **Ownership/control:** members can view their own data, correct errors, and request deletion at any time, and the data isn't passed to other parties without consent — giving them real control over data that is about them.

How the marks are awarded

- Transparency (2): clear statement of what/why/where/who BEFORE collection so members can give informed consent.
- Ownership & control (2): members can access, correct, withdraw and delete their data; it isn't handed on without consent.
- Answers must be specific to StepUp's fitness data, not generic "be honest".

Q4. Discuss · 6 marks

StepUp's analyst has only three months of data, from the 20% of members who use the app most. Management wants a public report claiming "StepUp members are the fittest in the suburb." **Discuss** the ethical issues raised by producing this report. ()

✗ A weak answer

The data is incomplete so the report is wrong. Also it's not transparent. They shouldn't do it because it's dishonest.

✓ A strong answer

Two issues stand out. First, **inaccurate/incomplete data**: the claim rests on three months of data from only the most active 20% of members — an unrepresentative sample. Presenting this as a firm fact about all members lets the public and prospective members be misled by a conclusion the data cannot support. Second, **transparency**: if the report hides the sample size and selection, readers can't judge how reliable the claim is. There is a tension — StepUp legally owns the data and may genuinely want to promote itself — but ethically it should disclose the limitations and avoid an overstated claim, because the people harmed are members and the public who trust the figure. The ethical path is to report honestly: "among our most active users, fitness improved," with the sample stated.

How the marks are awarded

- Identifies at least two relevant issues — typically inaccurate/incomplete data and transparency (misuse also creditable).
- For each: explains WHY it's an ethical issue (not just a quality label) and the impact on people (members, public).
- Shows the tension/both sides expected by "discuss" — e.g. business's right to promote vs duty to be honest.
- Reaches a reasoned position. 5–6 marks: two issues fully developed with stakeholders and a justified stance.